

Enstore

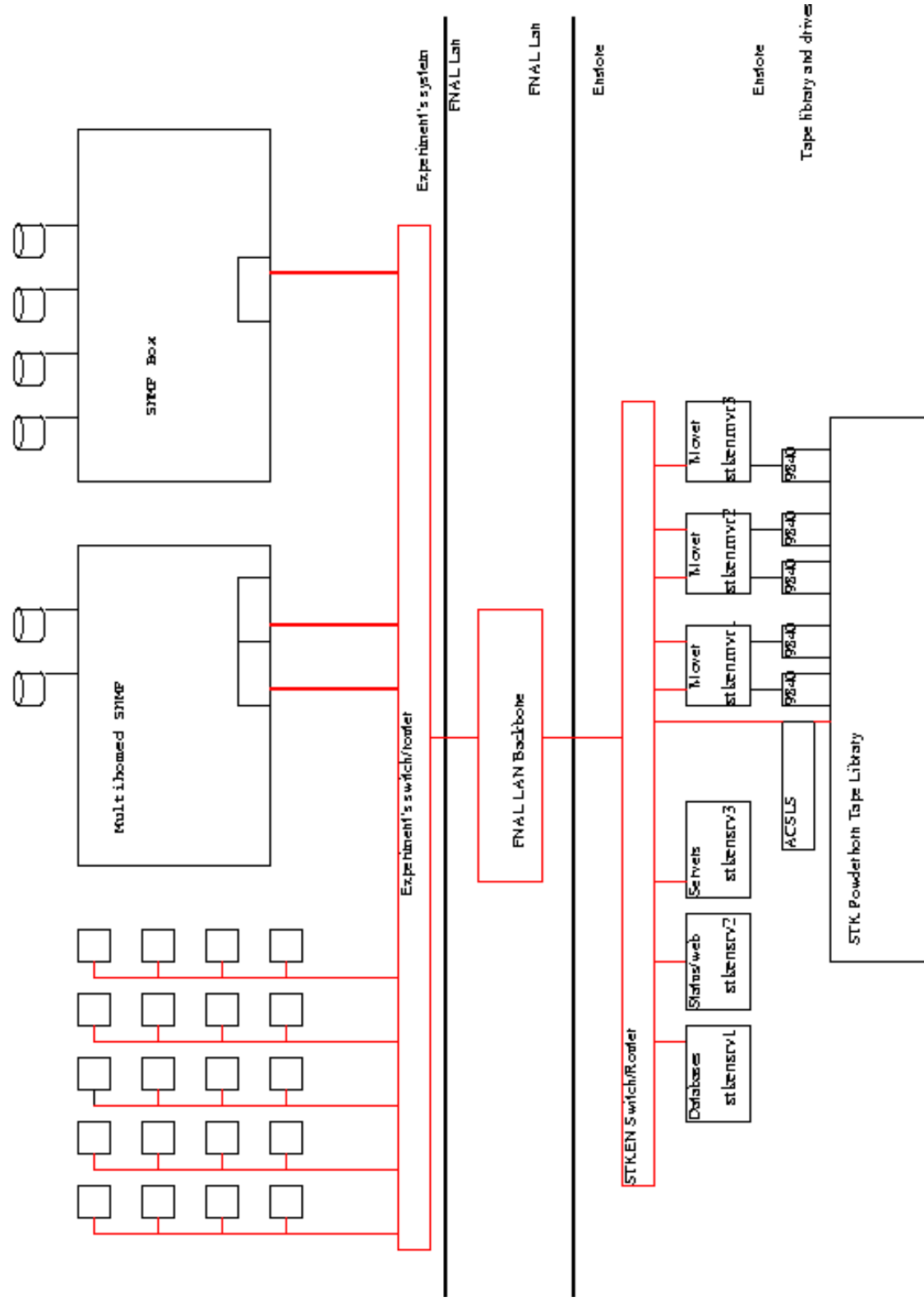
Service Envisioned

- Primary data store for experiment's large data sets, accessible via LAN
- High fault tolerance - ensemble reliability of a large tape drive plant, availability sufficient for DAQ
- Allow for automated tape libraries and manual tapes
- Catalog files in a distributed virtual filesystem (PNFS)
- Organize files efficiently on tape and expose information about tape organization to users
- CD will operate all the tape equipment
- Easy administration and monitoring

Hardware for Early Use

- 1 each - STK 9310 “Powderhorn” silo
- 5 each - STK 9840 “Eagle” tape drives
 - 11.5 MB/second
 - used at BaBar, CERN, RHIC
- 1500 - STK 9840 tape cartridges
 - 20 GB/ cartridge
- Linux Server and Mover computers
- FNAL standard network

Typical Overall System Incorporating Enstore



Service for First Users

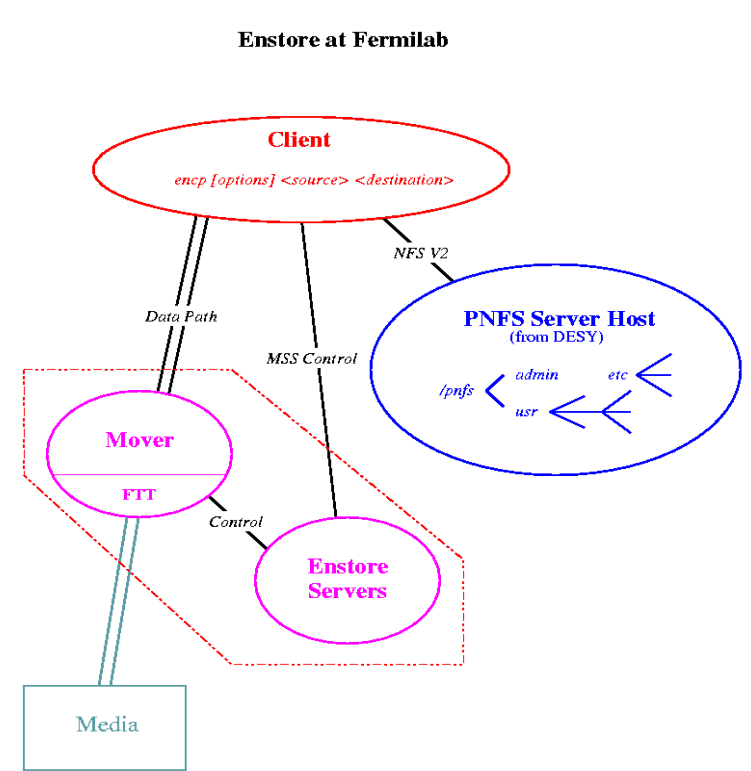
- Software in production (4 TB) for D0 Run II AML/2 tape library: 8MM, DLT drives
- STK system:
 - Only working days, working hours
 - Small data volumes ~1 TB for trial use
 - May have to upgrade LAN, network interfaces
 - Must be willing to point out bugs and problems
 - New hardware => small chance of data loss

Ease of Use

- Experiment can access tape as easily as a native file system
- PNFS mounted like NFS, viewable with UNIX commands (e.g. “ls”)
- Data is accessed with “encp” (Enstore Copy)
- Command syntax is similar to the UNIX “cp” command
 - `encp /scratch/qcd* /pnfs/theory/myproject`
- Delegate volume management to experiments
- Web-based monitoring

Basic Structure

- PNFS to name tape files using UNIX-like paths, served with NFS 2 transport
- Servers to schedule, configure, manage
- Movers to transport data and assure data integrity



Volume Principles

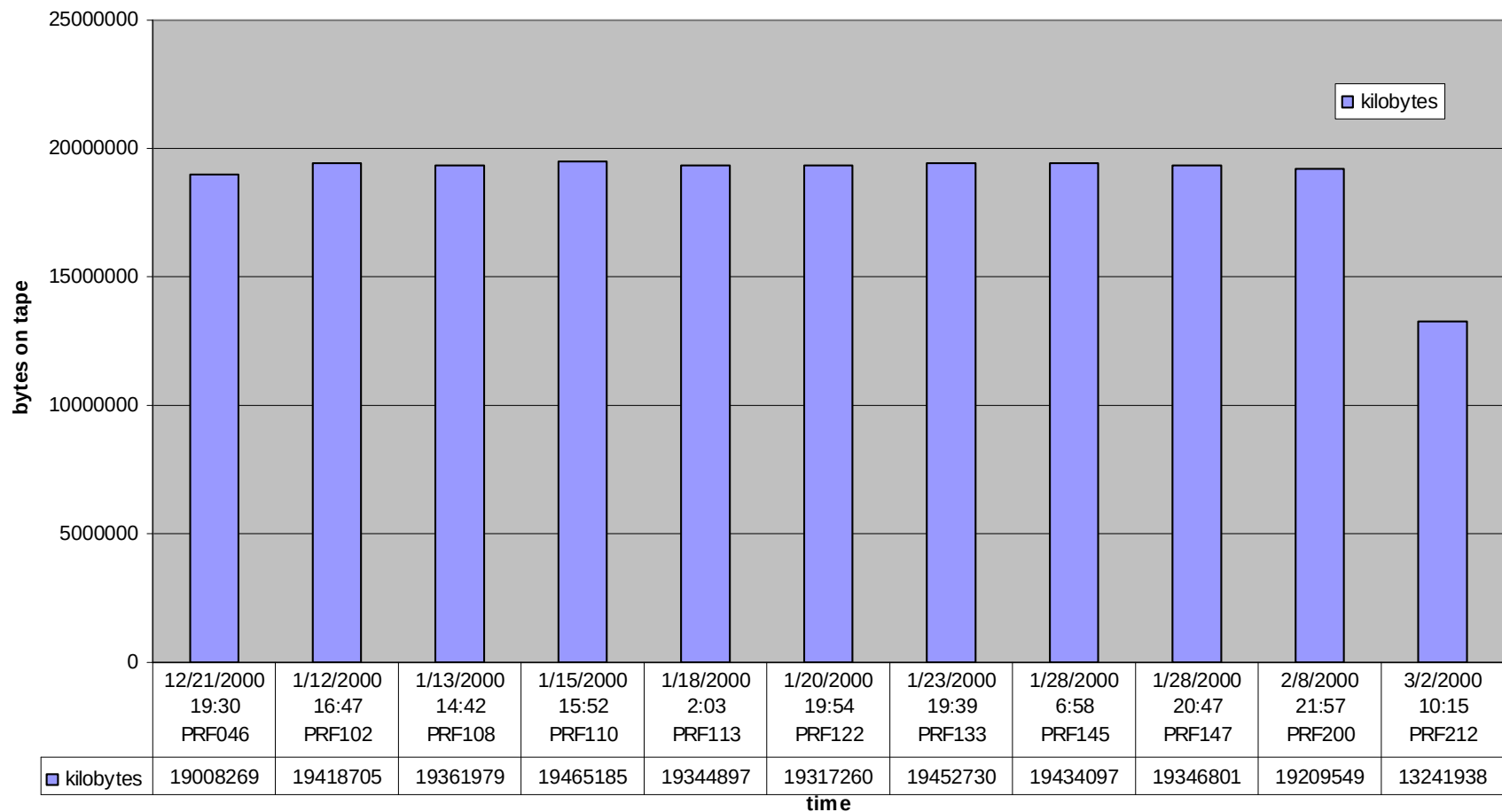
- Cluster related files on the same tapes
 - Enstore provides grouping primitives
- Allow for more tapes than robot can hold
 - Enstore provides quotas in tapes and robot “slots”
- Allow users to generate tapes (and metadata) outside our system
 - Enstore “volume import” provides tools to do this
- Allow tapes to leave our system and be readable with simple tools
 - Enstore can make tapes dumpable with cpio

Grouping on Tapes

- Grouping by category
 - “File families”: only files of the same family are on the same tape
 - A family is just an ASCII name
 - Names are administered by the experiment
- Grouping by time
 - Enstore closes volume for write when the next file does not fit
- Constrained parallelism
 - “Width” associated with a file family limits the number of volumes open for writing, concentrates files on fewer volumes
 - Higher width allows bandwidth into a file family to exceed the bandwidth of a tape drive

File family, width=1 over time

Time ordered, fully packed volumes



Tape Details

- Files do not stripe or span volumes
- Implementation details:
 - Tapes have ANSI VOL1 headers
 - Tapes are file structured as cpio archives, one file to an archive, one filemark per archive
- You can remove tapes from Enstore and just read them with GNU cpio (gives a 4GB limit right now)
- ANSI tapes planned, promised for D0

Library Manager

- Manages a homogeneous collection of tapes
- Accepts, queues and prioritizes encp client requests
- Dispatches work to Movers
- Will reorder work queues to minimize tape mounts and tape motion
- Identifies failed tape volumes

PNFS (DESY)

- Provides a directory tree to name files as you wish
- “Tags” provide per-directory defaults for creating files, which the experiment can administer (e.g. file family)
- “Layers” hold file metadata (e.g. volume label)
- Provides “volume map” directories where files are organized by physical location:
 - */pnfs/mountpoint/volmap/family/volume/*
 - Filenames of form *partition_block_filemark* (easily sortable)

PNFS:UNIX features

- Implemented using PNFS from DESY. NFS v2 “Transport”
- “Almost all” UNIX Utilities work, ls, find
- Standard utility reads/writes fail by design
- Many files in a directory is a poor choice “by design”
 - `$ pwd`
 - `/pnfs/sam/mammoth/mcc99_2/in2p3`
 - `[$ du -sk`
 - `267171544 .`
 - `$ ls -al sim.pmc02_in2p3.pythia.qcd_pt20.0_skip5800_mb1.1av_200evts.299_1138`
 - `-rw-r--r-- 1 sam root 250184748 Nov 30 17:25 sim.pmc02_in2p3.pythia.qcd_pt20.0_skip5800_mb1.1av_200evts.299_1138`
 - `rm sim.pmc02_in2p3.pythia.qcd_pt20.0_skip5800_mb1.1av_200evts.299_1138`
 - `rm: sim.pmc02_in2p3.pythia.qcd_pt20.0_skip5800_mb1.1av_200evts.299_1138: Permission denied`
 - `$ cat sim.pmc02_in2p3.pythia.qcd_pt20.0skip5800_mb1.1av_200evts.299_1138 > /dev/null`
 - `cat: sim.pmc02_in2p3.pythia.qcd_pt20.0_skip5800_mb1.1av_200evts.299_1138: Input/output error`

PNFS:defaults for new files

- Metadata “tags” are associated with directories
- Accessed by the “enstore pnfs” command.
- Inherited on “mkdir”.
- Initial tag on initial directory given by ISD dept
- Administered by the experiment

- [petravic@d0enmvr17a in2p3]\$ enstore pnfs --tags .
- .(tag)(library) = sammam
- .(tag)(file_family) = in2p3
- .(tag)(file_family_width) = 1
- .(tag)(file_family_wrapper) = cpio_odc

PNFS: File Metadata

- Describes an existing file
 - Accessed by the “enstore pnfs” command
 - Set by encp when the file is created
-
- `$ enstore pnfs --info sim.pmc02_in2p3.pythia.qcd_pt20.0_10000evts_skip5800_mb1.1av_200evts.299_1138`
 - `bfid="94400431100000L";`
 - `volume="PRF020";`
 - `location_cookie="0000_000005442_0000004";`
 - `size="250184748L";`
 - `file_family="in2p3";`
 - `map_file="/pnfs/sam/mammoth/volmap/in2p3/PRF020/0000_000005442_0000004";`

Removing Files

- Files may be removed using “rm”.
- User can scratch tape when all files on it are rm’ed. [enstore volume --delete]
- User can use a recovery utility to restore files up until the time the volume is scratched. [enstore file --restore]
- Files are recovered to pathname they were created with.

PNFS implementation (DESY)



- Mountd
- “nfsd”
- shared memory bkplane
- dbservers (gdbm)

System Admin's View of PNFS

- NFSv2
 - is the transport for the name space
 - gives authentication and authorization for access to metadata UID/GID + host based control on export
- Model FSTAB entry:
 - stkensrv1:/theory /pnfs/theory nfs user,intr,bg,hard,rw,noac 0 0
- user File data is NEVER transmitted via NFS
- NFS statistics at: <http://stkensrv2.fnal.gov:/enstore/pnfsRate.html>
- showmount is broken:
- hppc> showmount -e stkensrv1
- showmount: Can't decode result

http://stkenstrv2.fnal.gov:/enstore/pnfsRate.html

PNFS Counter Status Page - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Refresh Home Search Favorites History Mail Print Edit

Address <http://stkenstrv2.fnal.gov:/enstore/pnfsRate.html> Go Links >>

PNFS Counter Fetch Begin: Fri Aug 25 10:46:02 2000

Fri Aug 25 10:46:07 2000				Fri Aug 25 10:46:12 2000				Fri Aug 25 10:46:17 2000			
admin:0:r:enabled:/diska/pnfs/db/admin				eagle:1:r:enabled:/diska/pnfs/db/eagle				NULL:2:r:enabled:/diska/pnfs/db/NULL			
Attribute	Absolute	Change	Rate	Attribute	Absolute	Change	Rate	Attribute	Absolute	Change	Rate
-----	-----	-----	----	-----	-----	-----	----	-----	-----	-----	----
NULL	0	0	0/s	NULL	0	0	0/s	NULL	0	0	0/s
addtodir	1	0	0/s	addtodir	0	0	0/s	addtodir	0	0	0/s
admin	0	0	0/s	chparent	0	0	0/s	chparent	0	0	0/s
chparent	0	0	0/s	command	0	0	0/s	command	0	0	0/s
command	0	0	0/s	delobject	0	0	0/s	delobject	0	0	0/s
delobject	0	0	0/s	eagle	1	0	0/s	delobject	0	0	0/s
find_id	0	0	0/s	find_id	0	0	0/s	find_id	0	0	0/s
forcesize	0	0	0/s	forcesize	0	0	0/s	forcesize	0	0	0/s
get_chain	0	0	0/s	get_chain	0	0	0/s	get_chain	0	0	0/s
get_record	0	0	0/s	get_record	0	0	0/s	get_record	0	0	0/s
getattr	155813	3	0.6/s	getattr	10280	3	0.6/s	getattr	0	0	0/s
getroot	109909	0	0/s	getroot	0	0	0/s	getroot	0	0	0/s
lookup	58913	0	0/s	lookup	14652	0	0/s	lookup	0	0	0/s
looponly	769274	0	0/s	looponly	0	0	0/s	looponly	0	0	0/s
mkdir	0	0	0/s	mkdir	0	0	0/s	mkdir	0	0	0/s
mkfile	1	0	0/s	mkfile	6	0	0/s	mkfile	0	0	0/s
mlink	1	0	0/s	mlink	0	0	0/s	mlink	0	0	0/s
mod_link	0	0	0/s	mod_link	0	0	0/s	mod_link	0	0	0/s
readdata	112993	1	0.2/s	readdata	722	1	0.2/s	readdata	0	0	0/s
readdir	270	0	0/s	readdir	1176	0	0/s	readdir	0	0	0/s
readlink	9701	0	0/s	readlink	0	0	0/s	readlink	0	0	0/s
rename	0	0	0/s	rename	0	0	0/s	rename	0	0	0/s
rmdir	0	0	0/s	rmdir	0	0	0/s	rmdir	0	0	0/s
rmfile	0	0	0/s	rmfile	0	0	0/s	rmfile	0	0	0/s
rmfromdir	0	0	0/s	rmfromdir	0	0	0/s	rmfromdir	0	0	0/s
setattr	0	0	0/s	setattr	6	0	0/s	setattr	0	0	0/s

Done Internet

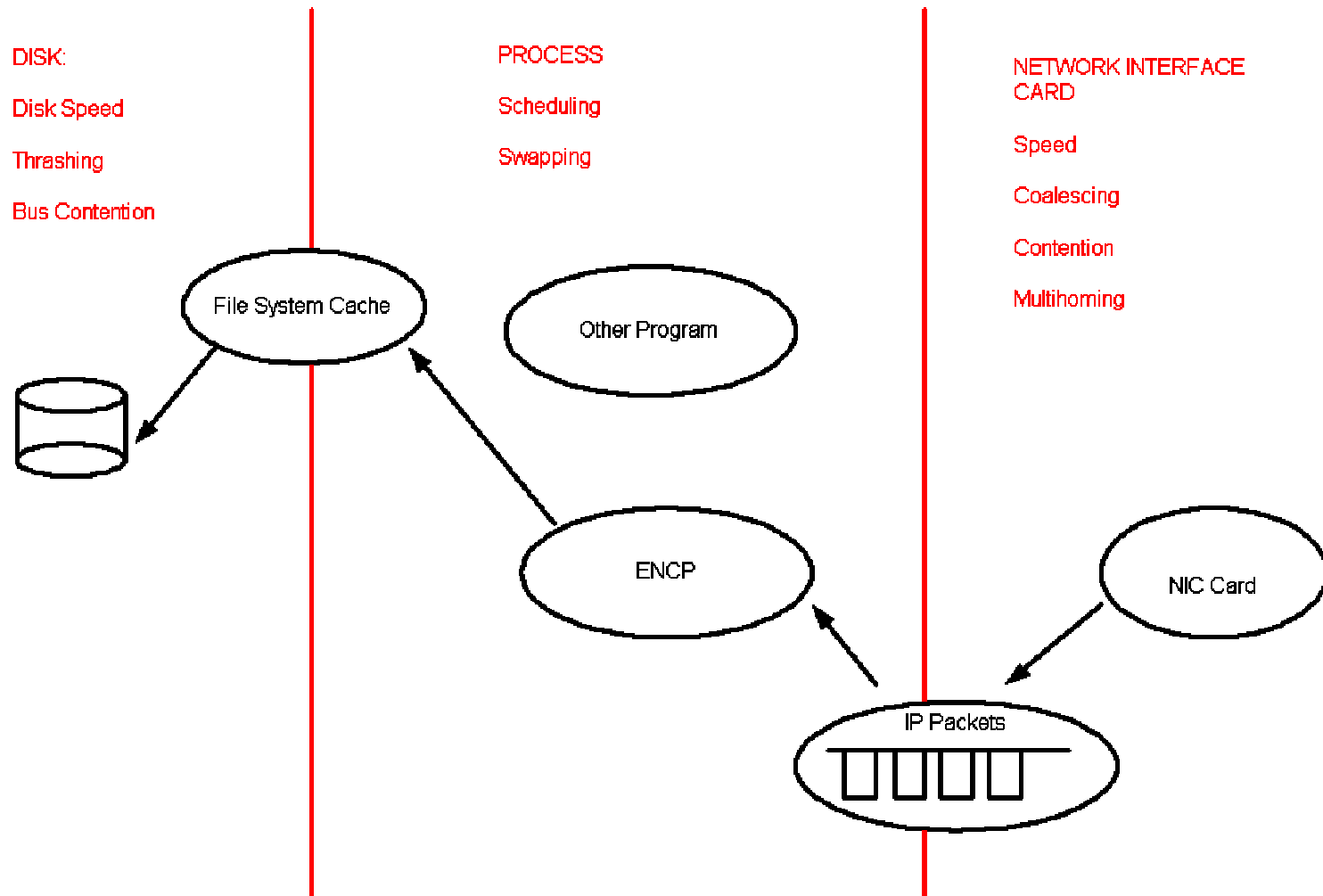
Sharing the Central Enstore System

- We make mount point(s) for your experiment
 - Host-based authentication on the server side for mounts.
 - Your meta data is in its own database files.
 - Under the mount point, UNIX file permission apply.
 - Make your uids/gid's uniform! (FNAL uniform UID/GIDS).
 - file permissions apply to the tag files as well.
- “Fair Share” envisioned, for tape drive resources.
 - Control over experiment resources by the experiment
- Priorities implemented for Data Acquisition.
 - Quick use of resources for the most urgent need

System Integration

- Hardware/system:
 - Consideration of upstream network.
 - Consideration of your NIC cards.
 - Good scheduling of the staging program.
 - Good throughput to your file systems.
- Software configuration
 - Software built for FUE platforms
 - Linux, IRIX, SunOS, OSF1

Elements of Good Throughput



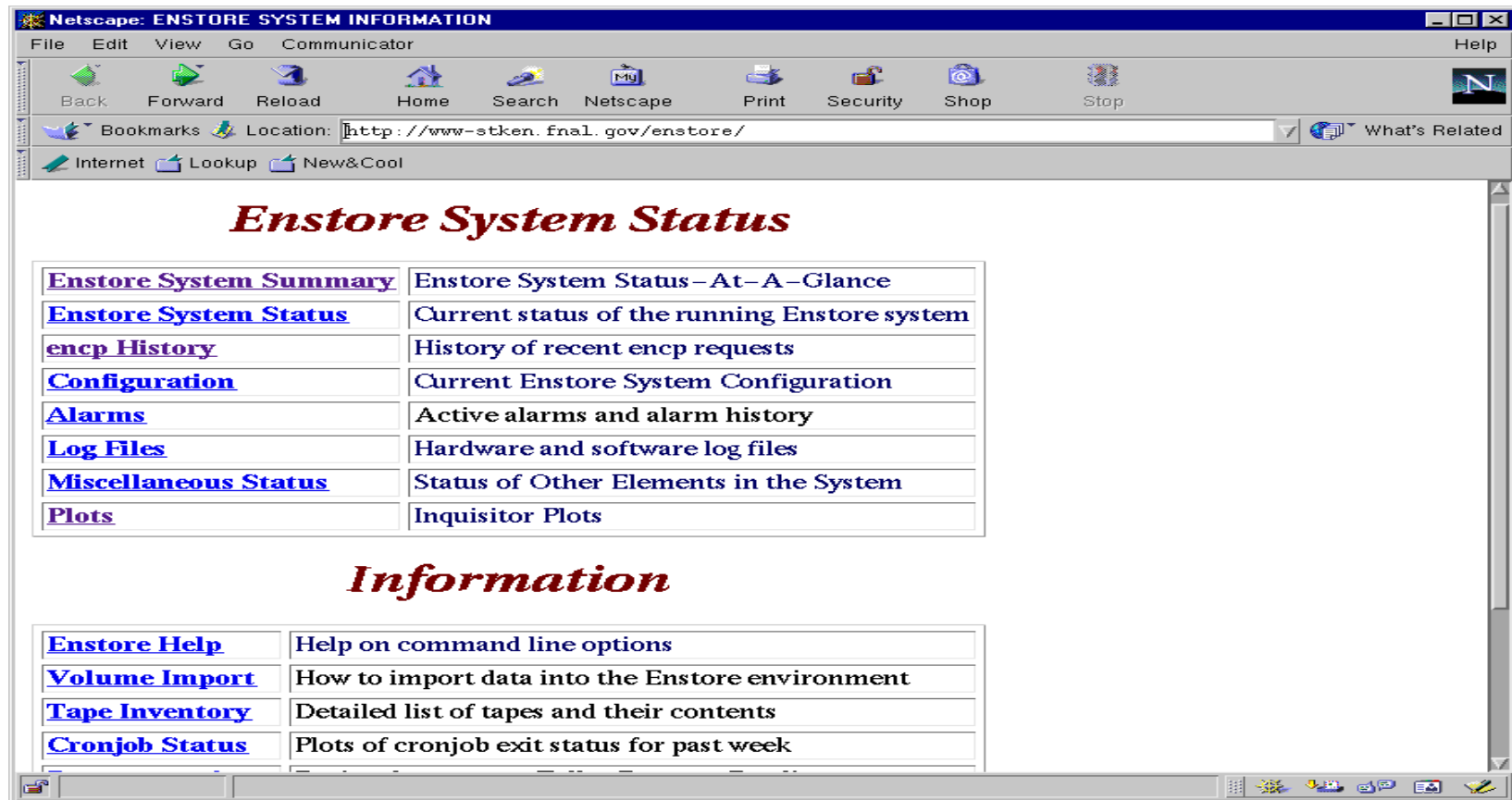
Checking....

- Enstore monitor:
 - <fsgi02> setup -q stken encp
 - <fsgi02> enstore monitor
 - Trying ('stkensrv1.fnal.gov', [], ['131.225.13.1']) Success. Network rate measured at 1.092 MB/S
 - <fsgi03> enstore monitor
 - Trying ('stkensrv1.fnal.gov', [], ['131.225.13.1']) Success. Network rate measured at 11.3 MB/S
- Recent rates available at: http://stkensrv2.fnal.gov/enstore/active_monitor.html
- Read a file to /dev/null
 - <fsgi03> encp --verbose=1 /pnfs/sdss/opdb/2000-Jun-16/d1/data1241_6_1.photo.DB /dev/null
 - Submitting read requests cumt= 0.399256944656
 - Q'd: /dev/null 961165828000000L bytes: 115146752 on VO0116 0000_0000000000_00000008 dt: 0.085179 cumt=0.485039
 - Receiving data for file /dev/null cumt= 0.611217021942
 - read_from_hsm /pnfs/sdss/opdb/2000-Jun-16/d1/data1241_6_1.photo.DB -> /dev/null: 115146752 bytes copied from VO0116 at 1.54 MB/S (8.68 MB/S) mover=eagle12.mover drive_id=9840 drive_sn=None drive_vendor=STK cumt=71.847064 {'media_changer': '....

<http://stkenstrv2.fnal.gov/enstore/>

- Source of interesting monitoring info
- Most updates are batched.
- Can see
 - recent transfers
 - the system is up or down
 - what transfers are queued
 - more

http://stkensrv2/enstore/



System Status: Green == Good

Netscape: ENSTORE Status-At-A-Glance

File Edit View Go Communicator Help

Back Forward Reload Home Search Netscape Print Security Shop Stop

Bookmarks Location: http://www-stken.fnal.gov/enstore/enstore_saag.html What's Related

Internet Lookup New&Cool

[Home](#) [System](#) [Servers](#) [Encp](#) [Help](#)

Enstore Status At-A-Glance

Brought To You By: SPAM
Last updated: 2000-Sep-12 14:40:19

Enstore Overall Status

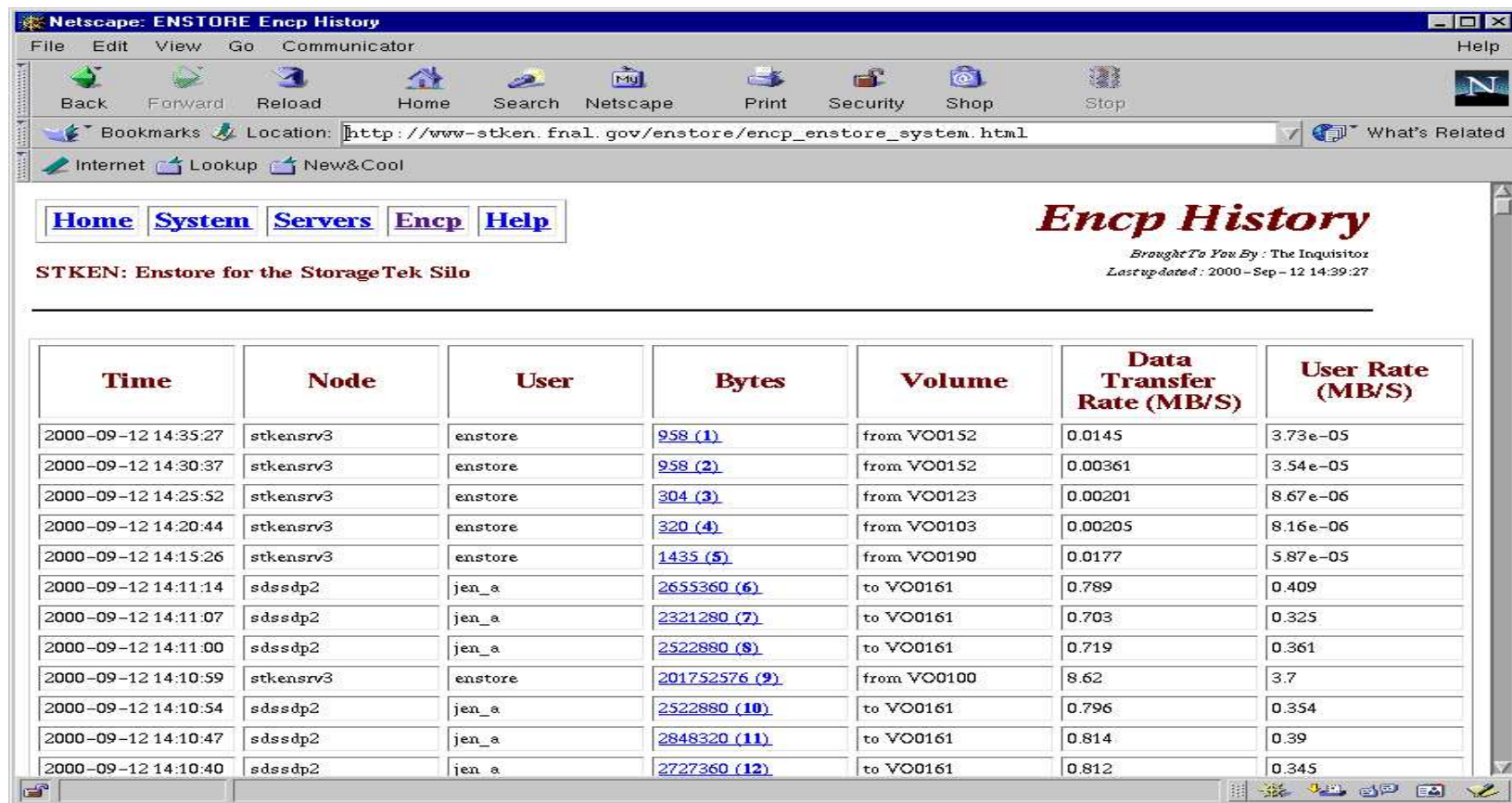
 [enstore](#)  [STK Powderhorn 9310](#)  [network](#)  [alarms](#)

Enstore Individual Server Status

<u>Servers</u>		<u>Library Managers</u>	<u>Media Changers</u>
 Alarm Server	 Configuration Server	 amlj-dlt.library_manager ✓ (Available, but not in production)	 STK.media_changer
 File Clerk	 Inquisitor	 eagle.library_manager	 amlj.media_changer ✓ (Available, but not in production)
 Logger	 Volume Clerk	 null1.library_manager	 null1.media_changer

<u>Movers</u>			
 JDE01.DLT.mover ✓ (Available, but not in production)	 JDE02.DLT.mover ✓ (Available, but not in production)	 JDE03.DLT.mover ✓ (Available, but not in production)	 JDE04.DLT.mover ✓ (Available, but not in production)
 eagle11.mover	 eagle12.mover	 eagle21.mover	 eagle22.mover
 eagle31.mover	 null41.mover	 null42.mover	 null51.mover

History of Recent Transfers

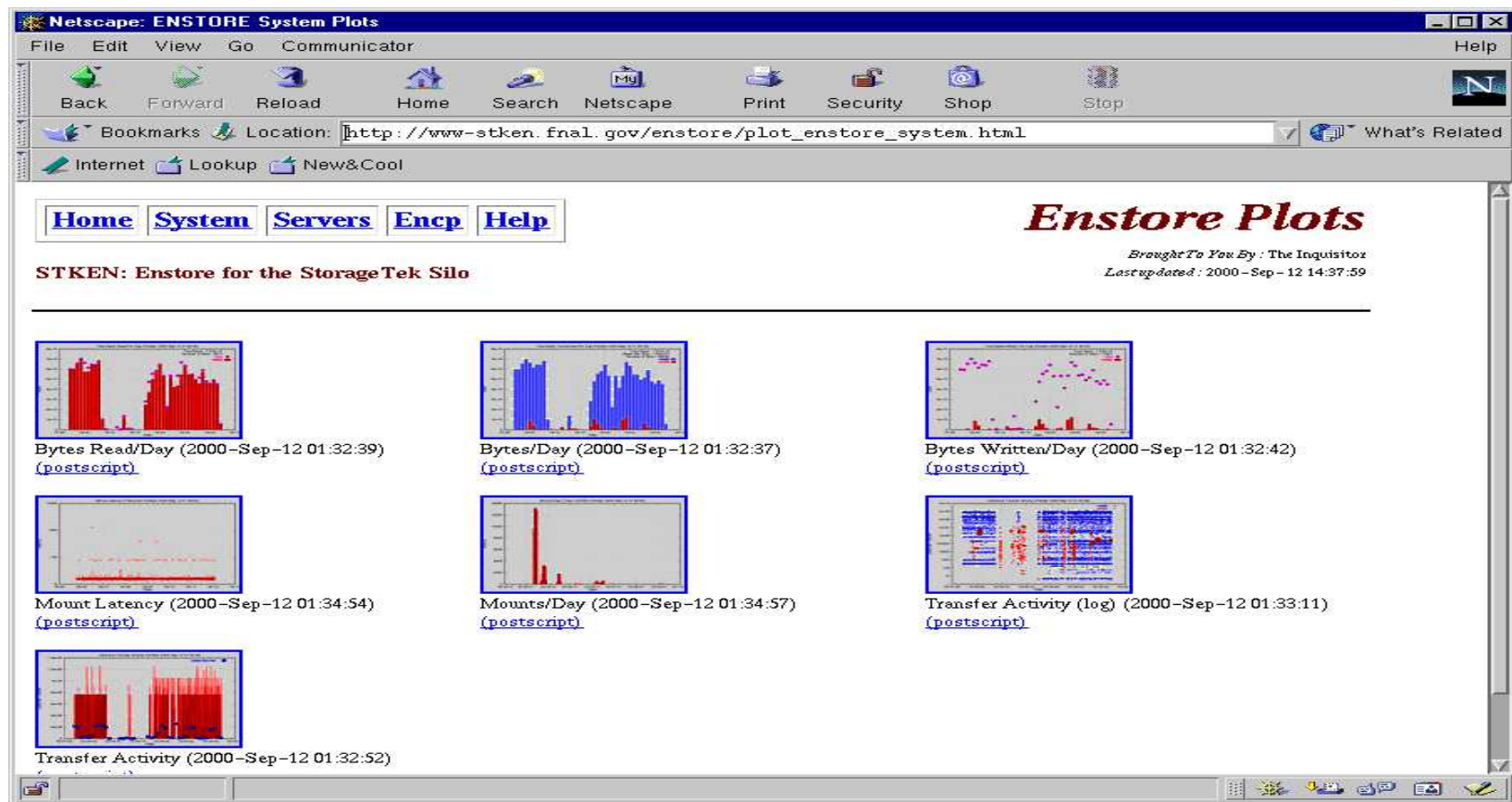


Encp History
Brought To You By: The Inquisitor
Last updated: 2000-Sep-12 14:39:27

STKEN: Enstore for the StorageTek Silo

Time	Node	User	Bytes	Volume	Data Transfer Rate (MB/S)	User Rate (MB/S)
2000-09-12 14:35:27	stkenstrv3	enstore	958 (1)	from VO0152	0.0145	3.73e-05
2000-09-12 14:30:37	stkenstrv3	enstore	958 (2)	from VO0152	0.00361	3.54e-05
2000-09-12 14:25:52	stkenstrv3	enstore	304 (3)	from VO0123	0.00201	8.67e-06
2000-09-12 14:20:44	stkenstrv3	enstore	320 (4)	from VO0103	0.00205	8.16e-06
2000-09-12 14:15:26	stkenstrv3	enstore	1435 (5)	from VO0190	0.0177	5.87e-05
2000-09-12 14:11:14	sdssdp2	jen_a	2655360 (6)	to VO0161	0.789	0.409
2000-09-12 14:11:07	sdssdp2	jen_a	2321280 (7)	to VO0161	0.703	0.325
2000-09-12 14:11:00	sdssdp2	jen_a	2522880 (8)	to VO0161	0.719	0.361
2000-09-12 14:10:59	stkenstrv3	enstore	201752576 (9)	from VO0100	8.62	3.7
2000-09-12 14:10:54	sdssdp2	jen_a	2522880 (10)	to VO0161	0.796	0.354
2000-09-12 14:10:47	sdssdp2	jen_a	2848320 (11)	to VO0161	0.814	0.39
2000-09-12 14:10:40	sdssdp2	jen_a	2727360 (12)	to VO0161	0.812	0.345

Summary Plots



Checklist to use Enstore

- Be authorized by the computing division.
- Identify performant disks and computers.
 - Use “bonnie” and “streams”
- Provide suitable network connectivity.
 - Use “enstore monitor” to measure.
- Plan use of namespace, file families.
- Regularize UIDs and GIDs if required.
- Mount the namespace.
- Use encp to access your files.

Support while using Enstore

Trouble reports to helpdesk

- Helpdesk@fnal.gov
- x2345

Information from us:

- Status pages on the web
- Stk-users@fnal.gov mailing list
 - We require at least one person from your group
- Very draft users guide.